

Yu-Hsiang Chen

Email ◇ GitHub ◇ LinkedIn ◇ Google Scholar ◇ Website

Education

National Yang Ming Chiao Tung University	Taiwan
M.Sc. in Computer Science	09/2024 – 06/2026
National Tsing Hua University	Taiwan
B.Sc. in Technology Management	09/2020 – 06/2024
◦ Ranked 2nd among 65 students every year. GPA 4.08/4.3.	

Research Experience

University of California, Berkeley	Berkeley, CA
Researcher, MSC Lab; supervised by Prof. Masayoshi Tomizuka	06/2025 – 12/2025
◦ Developed ECoSim , first data-efficient framework integrating diffusion and autoregressive models for controllable, realistic closed-loop traffic simulation; accepted at ECCV 2026.	
◦ Created HetroD , the first high-fidelity drone dataset and benchmark for evaluating heterogeneous traffic in autonomous driving; accepted at ICRA 2026.	
National Yang Ming Chiao Tung University	Hsinchu, Taiwan
Researcher, HCIS Lab; supervised by Prof. Yi-Ting Chen	09/2024 – 06/2026
◦ Developed NudgeVAD , a language-guided end-to-end driving framework using FiLM residuals; won 1st place in the DriveX DoScenes Challenge at CVPR Workshops 2026.	
◦ Co-developed frameworks for activation steering in end-to-end autonomous driving, safety validation, and VLM-based V2X scenario understanding, resulting in publications at IEEE ITSC 2026, ICCV Workshops 2025, and ECCV Workshops 2026.	
National Tsing Hua University	Hsinchu, Taiwan
Undergraduate Researcher; supervised by Prof. Hung-Min Sun	09/2022 – 06/2024
◦ Conducted NSTC-funded undergraduate research project on high-frequency trading; engineered a 400,000-record data pipeline and a campus application that improved operational efficiency by 45%.	

Industry Experience

Synology	Taipei, Taiwan
Research Intern	07/2026 – 02/2027
◦ Developed first generalizable tiny VLM for fast, accurate open-vocabulary attribute prediction on resource-constrained NAS edge devices; under review at CVPR 2027.	
Geniideas	Hsinchu, Taiwan
Research and Development Intern	06/2023 – 06/2026
◦ Developed industrial AI systems for automated piping ISO generation, laser-welding inspection, and crystal-oscillator classification for clients including TSMC and Micron.	

Selected Publications

ECoSim: Data-Efficient Fine-Tuning for Controllable Traffic Simulation	View
Y.-H. Chen , W.-J. Chang, Y.-T. Chen, and M. Tomizuka	
◦ ECCV 2026.	
HetroD: A High-Fidelity Drone Dataset and Benchmark for Heterogeneous Traffic in Autonomous Driving	View
Y.-H. Chen , W.-J. Chang, C. Kotulla, T. Keutgens, S. Runde, T. Moers, C. Klas, W. Zhan, M. Tomizuka, and Y.-T. Chen	
◦ ICRA 2026.	
NudgeVAD: Language-Nudged End-to-End Driving via FiLM Residuals	View
C.-C. Yang*, Y.-H. Chen *, and Y.-T. Chen	
◦ CVPR Workshops 2026 — 1st Place, DriveX DoScenes Challenge. *Co-first authors.	
V2X-Based Logical Scenario Understanding with Vision-Language Models	View
C.-L. Chi, Z.-H. Li, Y.-H. Chen , and Y.-T. Chen	
◦ ICCV Workshops 2025.	

Academic Service

DriveX and X-Sense Workshops	2025 – 2026
Organizer, ICCV 2025 and ECCV 2026	
◦ Launched the first Heterogeneous Traffic Simulation Challenge. View workshop	
Conference and Journal Reviewer	2025 – Present
NeurIPS, CVPR, ECCV, ICRA, IEEE RA-L, IEEE Transactions on Reliability, and IJPRAI	

Teaching Experience

National Yang Ming Chiao Tung University	Taiwan
Teaching Assistant for Introduction to Artificial Intelligence and Physical AI	2025-2026

Awards and Scholarships

1st Place, DriveX DoScenes Challenge , CVPR Workshops	2026
Five-time Presidential Award for Academic Excellence , National Tsing Hua University	2020 – 2024
Outstanding Graduate Award , National Tsing Hua University	2024
Ministry of Education Scholarship for Overseas Internship , Taiwan	2024
Undergraduate Research Scholarship , National Science and Technology Council	2023
Outstanding Intern Award , Hsinchu County Government	2023
General Education Competition Award , National Tsing Hua University	2023

Technical Skills

- **Programming Languages:** Python, C++, C, JavaScript
- **ML/AI:** Autonomous Driving, Computer Vision, Generative AI, Mathematical Statistics, PyTorch